Commentary on:
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Jazz Improvisation, Creativity, and Brain Plasticity

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In this issue of **WORLD NEUROSURGERY**, Galarza et al. report the amazing case of a professional jazz guitarist who underwent a left temporal lobectomy for an arteriovenous malformation (AVM), which elicited epilepsy and which finally bled. After a severe retrograde amnesia with a complete loss of musical interest and abilities, the patient recovered over time and has regained his previous virtuoso status. This is a fantastic report. Despite the lack of preoperative neurocognitive assessment and functional magnetic resonance imaging (fMRI), the extraordinary history of a high-level recovery in an excellent jazzman is very interesting. Indeed, the delayed neuropsychological examination correlated with anatomic MRI demonstrates the huge plastic potential of human brain in adults, including for musicality.

Neuroplasticity is a continuous process allowing short-term, middle-term, and long-term remodeling of neuronosynaptic maps, to optimize the functioning of cerebral networks (2). In normal brain, plasticity plays a critical role during phylogenesis as well as during ontogeny, with the elaboration of new circuits induced by learning, and the maintenance of neural networks in adults, then in elderly people: this is called the "natural plasticity" (3). Strong relationships seem to exist between music and plastic potential. In string players, an increased cortical representation of the digits has been demonstrated, suggesting that the representation of different parts of the body in the primary somatosensory cortex of humans, especially musicians, depends on use and changes to conform to the current needs and experience of the individuals (14).

Interestingly, beyond the movement itself, complex spontaneous musical performance involves a wider neural network. This is particularly true for improvisation in jazz performers, requiring artistic creativity in order to make immediate decisions about which musical phrases to invent and to play (25). In this state of mind, recent neuroimaging studies have shown that cerebral regions deactivated during improvisation were also at rest during dreaming and meditation, whereas activated areas included those controlling language and sensorimotor skills (22). Indeed, by employing 2 paradigms that differed widely in musical complexity, Limb et al. investigated improvisation in professional jazzmen using fMRI (21). They found that improvisation (compared to production of overlearned musical sequences) was consistently characterized by a dissociated pattern of activity in the prefrontal cortex using fMRI: extensive deactivation of dorsolateral prefrontal and lateral orbital regions with focal activation of the medial prefrontal (frontal polar) cortex. Moreover, changes in prefrontal activity during improvisation were accompanied by widespread activation of neocortical sensorimotor areas (that mediate the organization and execution of musical performance) as well as deactivation of limbic structures (that regulate motivation and emotional tone) (21). Therefore, because jazz improvisation may be one of the most useful experimental models for the study of spontaneous creativity, improvisation has been proposed as a new therapeutic tool within music therapy in neurologically impaired individuals (30). As a consequence, in the case reported by Galarza et al., even though fMRI was not performed, one could reasonably hypothesize that jazz

Key words
- Amnesia
- Amygdala
- Auditory-motor interrelations
- Brain
- Brain plasticity
- Cerebral arteriovenous malformation
- Epilepsy
- Hippocampus
- Memory
- Music
- Neurologic deficit
- Neuropsychological assessment
- Neuroscience
- Temporal lobe

Abbreviations and Acronyms
AVM: Arteriovenous malformation
fMRI: Functional magnetic resonance imaging

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improvisation participated in cerebral reorganization (even before AVM bleeding) and served as a cognitive rehabilitation following left temporal lobectomy.

Furthermore, it is now well known that a high degree of neuroplasticity can also be elicited by neurologic damages, especially in cases of congenital or slow-growing lesions (1, 4, 8). In AVM, it was already proposed that this chronic disease might lead to the development of compensatory network for language (18). For example, a bilateral temporal lobe activation of language was shown in patients presenting with left temporal lobe AVMs located in regions that should be involved in language function according to the classical models of cognitive neurosciences (19, 23). In the same vein, redistribution of expressive language function to the contralateral hemisphere in cases of AVM located in the left frontal language region has been observed, thus confirming the hypothesis of a possible contralateral reorganization of motor language behavior in this type of lesion (20). Nonetheless, in the observation by Galarza et al., the severe postoperative cognitive deficit experienced by the jazz guitarist supports the fact that his left hemisphere was still implied in language processes, despite the presence of an AVM. This is in agreement with the fact that plasticity mechanisms follow a hierarchical model, so that first there is an intrinsic reorganization within the injured areas (sign of favorable outcome); if this reshaping is not sufficient, then other regions implicated in the functional network are recruited in the ipsilateral hemisphere (close or more remote to the damaged area) and then in the contralateral hemisphere if necessary (10). On the basis of this model, even if no postsurgical fMRI was performed to map language and music functions, one might suggest a shift from a left dominant organization before cerebral hemorrhage (explaining the onset of retrograde amnesia as well as abolition of musical capabilities after surgery) toward a more distributed and delocalized organization of neural circuits allowing progressive recovery.

Indeed, the new science of connectomics has shown that massive functional reshaping is possible only if the subcortical white matter pathways have been preserved after brain injury (10). In other words, axonal connectivity is the major limitation of neuroplasticity (5, 17, 32). In the present case report, postsurgical anatomic MRI showed that the anterior part of the left inferior longitudinal fascicle and uncinate fascicle have been removed during temporal lobectomy. These 2 tracts are known to participate in the ventral semantic stream, with a relay at the level of the temporal pole between both fascicles (11, 24). It is worth noting that this indirect route can be compensated by a direct ventral stream subserved by the inferior fronto-occipital fascicle (12), which seems to be preserved in the patient because it runs above the sphenoidal horn. Interestingly, this pathway is thought to be involved in amodal (verbal and nonverbal) semantics as well as in noetic consciousness (13, 26). From the Greek "noos" which means "knowledge, mind," noetic consciousness is a level of consciousness peculiar to semantic memory. It is the awareness of knowing and understanding the world and the self, and the ability to be aware of such an awareness (31). Therefore, an acute damage (AVM bleeding followed by lobectomy) of the ventral route may explain why the patient experienced severe memory disturbances with consequences on music processes. On the other hand, preservation of the left inferior frontooccipital fascicle likely allowed the remarkable degree of recovery in this jazz guitarist. Of note, the anterior cortical terminations of this tract are represented by prefrontal and lateral orbital regions (28), that is, the areas implied in improvisation according to the recent fMRI investigation by Limb et al. (21). It seems logical that a perfect integrity of circuits underlying multimodal semantic processing and noetic consciousness has been preserved to give access to a high level of creativity.

To conclude, the neuroscience of music improvisation could have strong implications in the better comprehension of mechanisms enabling reorganization of circuits that subserve high-order functions such as supramodal semantics and creativity. In this setting, this amazing case report supports the fact that distributed neural pattern may provide a cognitive context that allows the emergence of spontaneous creative activity. Of note, jazz improvisation with other musicians may also imply interactions with subcircuit underpinning the assessment of the subjective experience of others in terms of mental status, a brain function referred to as mentalizing (16). The improved knowledge of neural foundations underlying these complex functions may be very useful for developing a prophylactic brain surgery in chronic diseases, for example, AVM or slow-growing tumors like diffuse low-grade gliomas (7, 9). The aim would be to perform a preventive resection before the onset of neurologic worsening by using intraoperative cortical and subcortical mapping in awake patients, and by adapting the intrasurgical functional tasks according to the quality of life of each individual (6, 15). This personalized management means that beyond movement and language, new tests should be built and administrated in musicians—as previously reported for integrated functions as dancing (29), cross-modal judgment (27), or multimodal semantics (26). New specific programs of cognitive rehabilitation, possibly based on music therapy, could also be developed to potentiate cerebral plasticity and to optimize the postoperative neurologic recovery.

REFERENCES

1. Desmurget M, Bonnetblanc F, Duffau H: Contrasting acute and slow growing lesions: a new door to brain plasticity. Brain 130:898-914, 2007.
2. Duffau H: Lessons from brain mapping in surgery for low-grade glioma: insights into associations between tumour and brain plasticity. Lancet Neurol 4:476-486, 2005.
3. Duffau H: Brain plasticity: from pathophysiological mechanisms to therapeutic applications. J Clin Neurosci 13:885-897, 2006.
4. Duffau H: Brain plasticity and tumors. Adv Tech Stand Neurosurg 3:3-33, 2008.
5. Duffau H: Does post-lesional subcortical plasticity exist in the human brain? Neurosci Res 65:131-135, 2009.
6. Duffau H: Awake surgery for nonlanguage mapping. Neurosurgery 66:523-529, 2010.
7. Duffau H: Awake surgery for incidental WHO grade II gliomas involving eloquent areas. World Neurosurg 154:575-584, 2012.
8. Duffau H: The "frontal syndrome" revisited: lessons from electrostimulation mapping studies. Cortex 48:120-131, 2012.
9. Duffau H: The rationale to perform early resection in incidental diffuse low-grade gliomas: toward a "preventive surgical neurooncology." World Neurosurg 2012 Jun 25 [Epub ahead of print].
10. Duffau H: The huge plastic potential of adult brain and the role of connectomics: new insights provided by serial mappings in glioma surgery. Cortex 2013 Aug 19 [Epub ahead of print].

11. Duffau H, Gatignol P, Moritz-Gasser S, Mandonnet E: Is the left uncinate fasciculus essential for language? A cerebral stimulation study. J Neurol 256:382-389, 2009.

12. Duffau H, Herbet G, Moritz-Gasser S: Toward a pluri-component, multimodal, and dynamic organization of the ventral semantic stream in humans: lessons from stimulation mapping in awake patients. Front Syst Neurosci 7:44, 2013.

13. Duffau H, Moritz-Gasser S, Mandonnet E: A re-examination of neural basis of language processing: proposal of a dynamic hodotopical model from data provided by brain stimulation mapping during picture naming. Brain Lang 2013 Jul 15 [Epub ahead of print].

14. Elbert T, Pantev C, Wienbruch C, Rockstroh B, Taub E: Increased cortical representation of the fingers of the left hand in string players. Science 270:305-307, 1995.

15. Fernández Coello A, Moritz-Gasser S, Martino J, Martinoni M, Matsuda R: Duffau H: Selection of intraoperative tasks for awake mapping based on relationships between tumor location and functional networks. J Neurosurg 2013 Sep 20 [Epub ahead of print].

16. Herbet G, Lafargue G, Bonnetblanc F, Moritz-Gasser S, Duffau H: Is the right frontal cortex really crucial in the mentalizing network? A longitudinal study in patients with a slow-growing lesion. Cortex 2013 Aug 14 [Epub ahead of print].

17. Ius T, Angelini E: Thiebaut de Schotten M, Mandonnet E, Duffau H: Evidence for potentials and limitations of brain plasticity using an atlas of functional resectability of WHO grade II gliomas: towards a "minimal common brain." Neuroimage 56:992-1000, 2011.

18. La Piana R, Klein D, Cortes MDP, Tampieri D: Speech reorganization after an AVM bleed cured by embolization. A case report and review of the literature. Interv Neuroradiol 15:456-461, 2009.

19. Latchaw RE, Hu X, Ugurbil K, Hall WA, Madison MT, Heros RC: Functional magnetic resonance imaging as a management tool for cerebral arteriovenous malformations. Neurosurgery 37:619-626, 1995.

20. Lazar RM, Marshall RS, Pile-Spellman J, Duong HC, Mohr JP, Young WL, Solomon RL, Perera GM, DeLaPaz RL: Interhemispheric transfer of language in patients with left frontal cerebral arteriovenous malformation. Neuropsychologia 38: 1325-1332, 2000.

21. Limb CJ, Braun AR: Neural substrates of spontaneous musical performance: an fMRI study of jazz improvisation. PLoS One 3:e1679, 2008.

22. López-González M, Limb CJ: Musical creativity and the brain. Cerebrum 2:2012, 2012.

23. Maldjian J, Atlas SW, Howard RS 2nd, Greenstein E, Alsop D, Detre JA, Listerud J, D'Esposito M, Flamm ES: Functional magnetic resonance imaging of regional brain activity in patients with intracerebral arteriovenous malformations before surgical or endovascular therapy. J Neurosurg 84:477-483, 1996.

24. Mandonnet E, Nouet A, Gatignol P, Capelle L, Duffau H: Does the left inferior longitudinal fasciculus play a role in language? A brain stimulation study. Brain 130:623-629, 2007.

25. McPherson M, Limb CJ: Difficulties in the neuroscience of creativity: jazz improvisation and the scientific method. Ann N Y Acad Sci 2013 Jul 19 [Epub ahead of print].

26. Moritz-Gasser S, Herbet G, Duffau H: Mapping the connectivity underlying multimodal (verbal and non-verbal) semantic processing: a brain electrostimulation study. Neuropsychologia 51: 1814-1822, 2013.

27. Plaza M, Gatignol P, Cohen H, Berger B, Duffau H: A discrete area within the left dorsolateral prefrontal cortex involved in visual-verbal incongruence judgment. Cereb Cortex 18: 1253-1259, 2008.

28. Sarubbo S, De Benedictis A, Maldonado IL, Basso G, Duffau H: Frontal terminations for the inferior fronto-occipital fascicle: anatomical dissection, DTI study and functional considerations on a multi-component bundle. Brain Struct Funct 218:21-37, 2013.

29. Thiebaut de Schotten M, Urbanski M, Duffau H, Volle E, Levy R, Dubois B: Bartolomeo P: Direct evidence for a parietal-frontal pathway subserving spatial awareness in humans. Science 309: 2226-2228, 2005.

30. Tomaino CM: Creativity and improvisation as therapeutic tools within music therapy. Ann N Y Acad Sci 2013 Aug 14 [Epub ahead of print].

31. Tulving E: Multiple memory systems and consciousness. Hum Neurobiol 6:67-80, 1987.

32. van Geemen K, Herbet G, Moritz-Gasser S, Duffau H: Limited plastic potential of the left ventral premotor cortex in speech articulation: evidence from intraoperative awake mapping in glioma patients. Hum Brain Mapp 2013 Apr 24 2013 Apr 24 [Epub ahead of print].

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